21. $\frac{d}{dx}(5e^x - 2\log x)$ is equal to

(a) $5e^x - \frac{2}{x}$

(b) $5e^x - 2x$

(c) $5e^x - \frac{1}{x}$

(d) $2\log x$

22. If $y = x$ and $z = \frac{1}{x}$,then $\frac{dy}{dz}$ =

(a) x^2 (b) 1

(c) $-x^2$ (d) $-\frac{1}{x^2}$

23. If $y = e^{2x}$,then $\frac{d^2y}{dx^2}$ at $x = 0$ is

(a) 4 (b) 9

(c) 2 (d) 0

24. If $y = \log x$,then y_2 =

(a) $\frac{1}{x}$ (b) $-\frac{1}{x^2}$

(c) $-\frac{2}{x^2}$ (d) e^2

25. $\frac{d}{dx}(a^x)$ =

(a) $\frac{1}{x\log_e a}$ (b) a^a

(c) $x\log_e a$ (d) $a^x \log_e a$

Miscellaneous Problems

1. If $f(x) = \frac{1}{2x+1}$, $x > -\frac{1}{2}$, then show that $f(f(x)) = \frac{2x+1}{2x+3}$.

2. Draw the graph of $y = 9 - x^2$

3. If $f(x) = \begin{cases} \frac{x-|x|}{x} & \text{if } x \neq 0 \\ 2 & \text{if } x = 0 \end{cases}$ then show that $\lim_{x\to 0} f(x)$ does not exist.

4. Evaluate: $\lim_{x\to 1} \frac{(2x-3)(\sqrt{x}-1)}{2x^2+x-3}$

5. Show that the function $f(x) = 2x - |x|$ is continuous at $x = 0$

6. Verify the continuity and differentiability of $f(x) = \begin{cases} 1-x & \text{if } x < 1 \\ (1-x)(2-x) & \text{if } 1 \le x \le 2 \\ 3-x & \text{if } x > 2 \end{cases}$ at $x = 1$ and $x = 2$

7. If $x^y = y^x$, then prove that $\frac{dy}{dx} = \frac{y}{x}\left(\frac{x\log y - y}{y\log x - x}\right)$

8. If $xy^2 = 1$, then prove that $2\frac{dy}{dx} + y^3 = 0$.

9. If $y = \tan x$, then prove that $y_2 - 2yy_1 = 0$.

10. If $y = 2\sin x + 3\cos x$, then show that $y_2 + y = 0$.

Summary

- Let A and B be two non empty sets, then a function f from A to B, associates every element of A to an unique element of B.
- $\lim_{x\to a} f(x)$ exists $\Leftrightarrow \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$.
- For the function $f(x)$ and a real number a, $\lim_{x\to a} f(x)$ and $f(a)$ may not be the same.
- A function $f(x)$ is a continuous function at $x = a$ if only if $\lim_{x\to a} f(x) = f(a)$.
- A function $f(x)$ is continuous, if it is continuous at every point of its domain.
- If $f(x)$ and $g(x)$ are continuous on their common domain, then $f \pm g$, $f \cdot g$, kf (k is a constant) are continuous and if $g \neq 0$ then $\frac{f}{g}$ is also continuous.

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